VIETNAM NATIONAL UNIVERSITY - HO CHI MINH CITY
HO CHI MINH CITY UNIVERSITY OF TECHNOLOGY
FACULTY OF COMPUTER SCIENCE AND ENGINEERING

MACHINE LEARNING

Assignment Report

Prediction with Traditional and Deep Learning Pipelines

Instructor: Truong Vinh Lan
Students: Phuong Xuong Duc - 2352270
Nguyen Tien Minh - 2352755
Tan Khanh Phong - 2352911
Nguyen Bui Huynh Khuong - 2352637
Ho Hong Phuc Nguyen - 2352824

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Task Distribution

No.	Full Name	Student ID	Assigned Tasks	Contribution
1	Phuong Xuong Duc	2352270	Exploratory Data Analysis	100%
2	Nguyen Tien Minh	2352755	Research and add more options for preventing overfitting, Summary and model comparison analysis	100%
3	Tan Khanh Phong	2352911	Data collection and feature pre processing pipeline.	100%
4	Nguyen Bui Huynh Khuong	2352637	Implementation of modern deep learning models and Write report	100%
5	Ho Hong Phuc Nguyen	2352824	Implementation of traditional machine learning models and Write report	100%

Table 1: Workload Distribution among Team Members

1 Introduction

1.1 Research Motivation and Objectives

Predicting an individual's income level based on census demographics is a foundational task in computational socio-economics. The ability to accurately classify whether a person's income exceeds a specific threshold provides critical insights into wealth distribution and systemic demographic disparities. The primary objective of this project is to architect a robust machine learning framework capable of high-precision income classification. Beyond predictive accuracy, this research aims to conduct a comparative benchmarking between traditional statistical models and modern deep learning architectures. Specifically, the project evaluates the trade-off between manual dimensionality reduction (via PCA) and end-to-end representation learning, aligning with the core academic goal of implementing a full-stack machine learning pipeline for tabular data.

1.2 Dataset Overview: The UCI Adult Census

This project utilizes the "Adult Census Income" dataset, a benchmark corpus sourced from the UCI Machine Learning Repository, originally extracted from the 1994 United States Census Bureau database. The dataset encapsulates 32,561 observations characterized by a heterogeneous feature space of 15 attributes, including age, education-num, occupation, marital status, race, sex, and native-country. The binary target variable classifies individuals into two economic tiers: $\leq 50K$ and $> 50K$. This dataset was strategically selected due to its complex mix of continuous and high-cardinality categorical variables, providing a rigorous environment for testing advanced encoding and standardization techniques.

1.3 Technical Challenges and Constraints

Implementing an effective model on this dataset involves navigating several critical constraints identified during initial analysis:

- **Structural Class Imbalance:** A primary challenge is the severe skewness in class distribution (75.9% vs 24.1%). This ratio renders global accuracy a misleading metric, necessitating a transition toward F1-Score and ROC-AUC evaluation.
- **High-Dimensional Sparsity:** The transformation of categorical features via *One-Hot Encoding* induces the "Curse of Dimensionality," creating a sparse matrix that can destabilize traditional distance-based classifiers.

- **Administrative Noise:** The raw census data contains administrative features (statistical population weights) that do not possess individual predictive power, requiring careful feature selection to prevent model bias.
- **Algorithmic Fairness:** Given the 1994 socio-political context, the dataset contains inherent historical biases regarding gender and race. Managing these biases is both a technical and ethical constraint.

1.4 Proposed Contribution

This report delivers a comprehensive, end-to-end implementation of a bifurcated machine learning pipeline. *Branch A* demonstrates the effectiveness of Principal Component Analysis (PCA) in compressing the feature space to optimize Logistic Regression (LR), Support Vector Machine (SVM), and Random Forest (RF). Simultaneously, *Branch B* implements a Modern Pipeline utilizing a Multi-layer Perceptron (MLP) trained on raw, preprocessed features to evaluate the latent representation learning capability of neural networks.

1.5 Code Availability and Reproducibility

To uphold the principles of open science and ensure the reproducibility of the experimental results presented in this report, the complete technical implementation has been made publicly available. This includes the end-to-end preprocessing pipeline, model architectures, and evaluation logs. The version-controlled repository, containing the modular Python source code and environment configurations, is hosted on GitHub. Furthermore, for immediate execution and verification, an interactive Google Colab notebook is provided, allowing for independent benchmarking of both the traditional and deep learning branches.

2 Exploratory Data Analysis (EDA)

The EDA phase was executed with rigorous statistical scrutiny to decode the underlying topological structure, latent socioeconomic biases, and inherent anomalies within the "Adult Census Income" dataset. This phase transcends mere descriptive statistics by mapping every empirical observation to a definitive "Actionable Insight," scientifically justifying the architectural configuration of the subsequent machine learning pipeline.

2.1 Data Topology and Structural Integrity Audit

A foundational audit was conducted to assess the feature taxonomy and memory footprint of the corpus $(32,561 \times 15)$.

- **Data Integrity:** 24 duplicated instances ($< 0.1\%$) were deterministically purged to ensure homoscedasticity and prevent "Data Leakage" risks, where models artificially inflate performance metrics by memorizing exact overlapping vectors.
- **Pruning Administrative Noise:** The "Statistical Population Weights" (`fnlwgt`) were permanently excluded. As these represent administrative scaling factors assigned by the Census Bureau rather than intrinsic personal attributes, they introduce survey-specific variance that is non-predictive for individual income tiers.

2.2 Missing Value Topology (MNAR Detection)

Missing data, encoded as "?", is exclusively localized within categorical attributes: `occupation` (5.66%), `workclass` (5.64%), and `native-country` (1.79%).

Actionable Insight: The near-perfect overlap between `workclass` and `occupation` missingness indicates a **Missing Not At Random (MNAR)** pattern, typically reflecting populations structurally outside the formal labor market or in transition. To preserve statistical power without introducing significant bias, a *SimpleImputer(strategy='most_frequent')* is deployed to stabilize the matrix while preserving its macroscopic distribution.

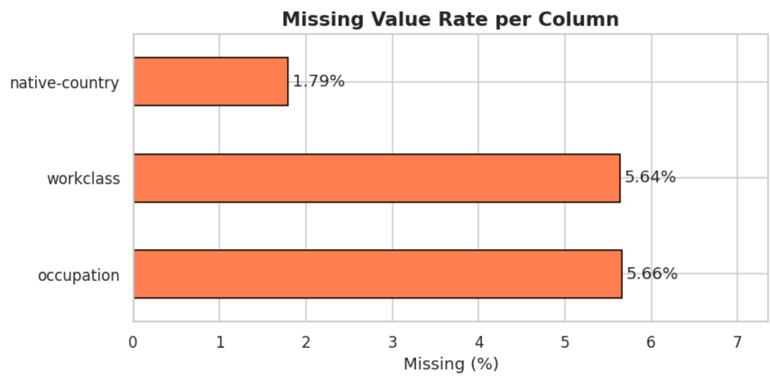

Figure 1: Analysis of missing values highlighting MNAR patterns in employment features.

2.3 Target Variable Anatomy: The Imbalance Problem

The `income` target exhibits a severe 3:1 imbalance ($75.9\% \leq 50K$ vs $24.1\% > 50K$).

Statistical Constraint: This distribution induces the **"Accuracy Paradox"**, where global accuracy becomes a misleading metric. Consequently, the evaluation framework prioritizes **F1-Score** and **ROC-AUC** to ensure sensitivity toward the minority class. Furthermore, **Stratified Sampling** is strictly enforced during the partitioning phase to prevent catastrophic class-starvation in training folds.

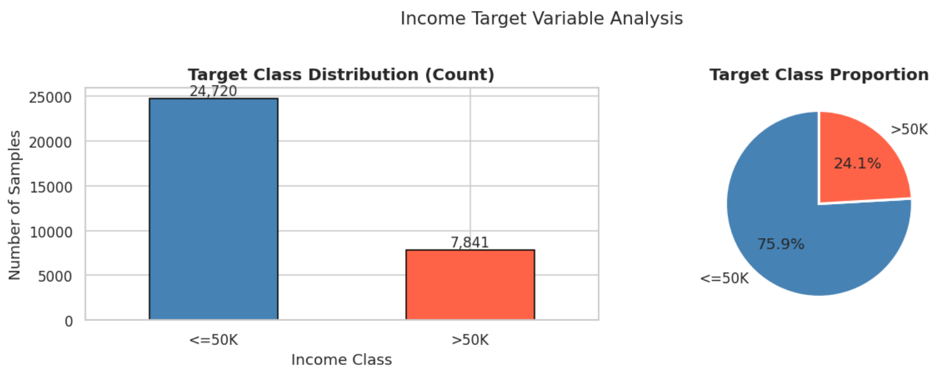

Figure 2: Class imbalance visualization: 75.9% majority vs 24.1% minority.

2.4 Univariate Dynamics: Continuous Feature Analysis

Age Progression and Earning Power: The distribution confirms income as a function of career maturity. The $> 50K$ class peaks significantly later (ages 37-50) than the $\leq 50K$ group (ages 19-30). Valid outliers (e.g., age 90) are retained to preserve the real-world socioeconomic variance inherent in the census data.

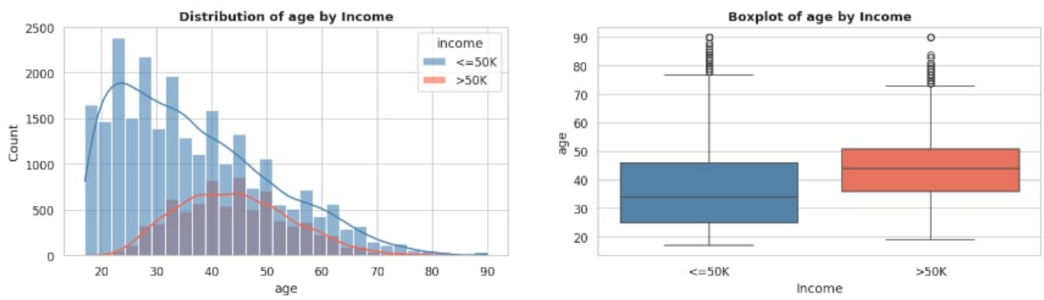

Figure 3: Age density trajectories reflecting human capital accumulation over time.

The Education Premium (Education-num): A strict monotonic correlation exists where higher numerical mappings shift the probability mass toward the high-income tier. Significant peaks at 9 (HS-grad), 10 (Some-college), and 13 (Bachelors) represent critical socioeconomic thresholds that act as primary decision boundaries for the estimators.

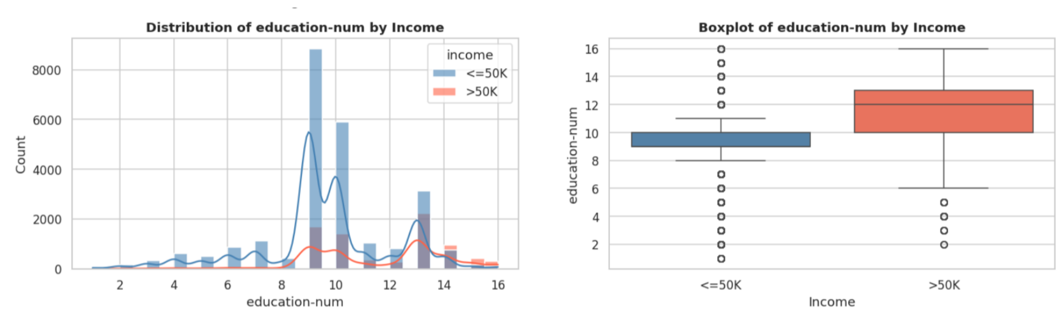

Figure 4: Education-num distribution: Higher numerical mappings as primary predictive drivers.

Capital Variance and Zero-Inflation: `Capital-gain` and `capital-loss` exhibit extreme right-skewness and over 91% zero-inflation. This sparsity confirms that significant financial investment is concentrated within a microscopic demographic subset. *Modeling Implication:* Such extreme variance destabilizes distance-based models (SVM) and gradient-based networks (MLP), making **StandardScaler** a mathematical prerequisite for convergence.

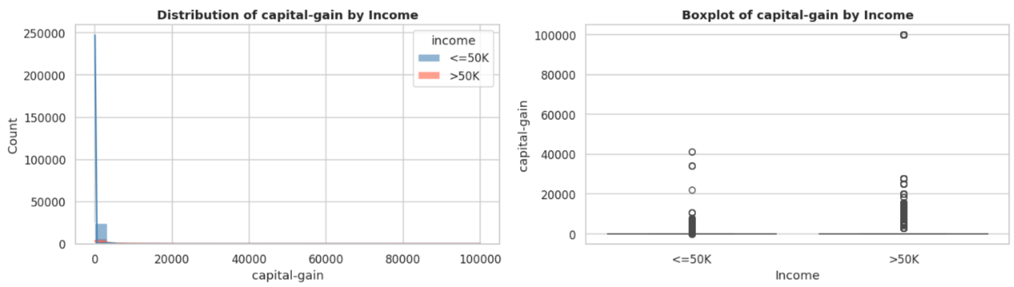

Figure 5: Capital-gain distribution: Extreme zero-inflation and its impact on variance.

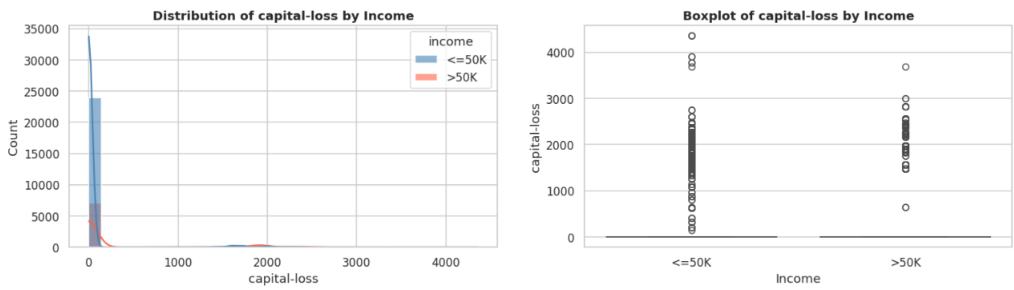

Figure 6: Capital-loss anatomy: High-earners exhibit a broader spectrum of investment risk.

Labor Intensity Trends (Hours-per-week): While 40 hours/week is the global mode, the $> 50K$ distribution is significantly broader, peaking in the 45-60 hour range. This validates labor intensity as a strong correlate of high income in the 1994 context.

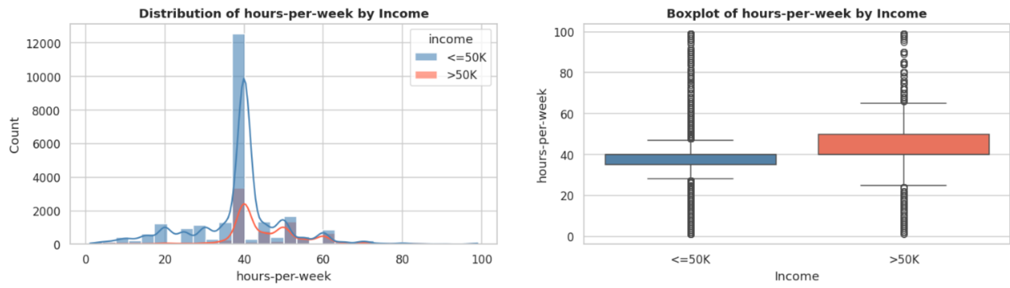

Figure 7: Labor intensity distribution: Contrasting standard vs. high-earner labor patterns.

2.5 Bivariate Dynamics: Socioeconomic & Demographic Biases

Professional and Educational Pillars: Bivariate analysis reveals that specialized sectors (Self-emp-inc) and advanced degrees (Doctorate, Prof-school) yield income rates $> 45\%$. This confirms the "Education Premium" as a dominant latent feature providing high information gain for classification.

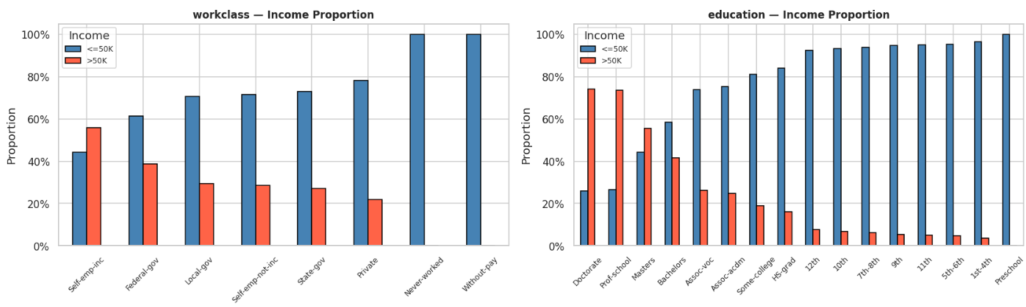

Figure 8: Socio-economic drivers: Impact of educational attainment and sector on income.

Social Stratification and Household Stability: Marital status serves as a powerful proxy for household stability, with "Married-civ-spouse" dominating the high-income bracket. Professionally, wealth is concentrated in "Exec-managerial" and "Prof-specialty" roles, while manual labor sectors show negligible high-income representation.

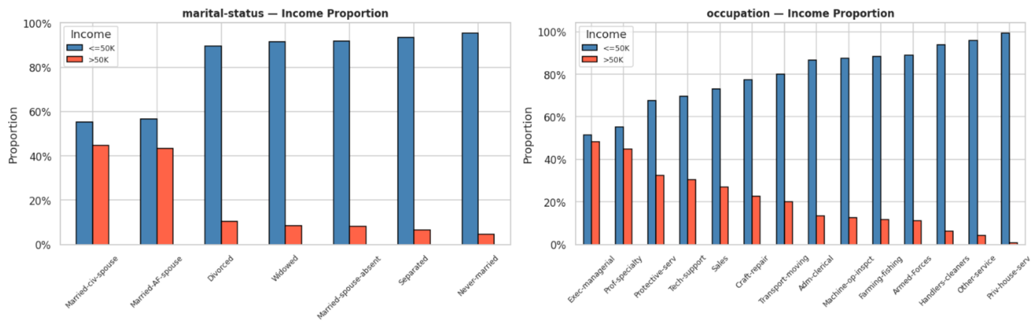

Figure 9: Social Stratification: Predictive influence of marital status and role specialization.

Household Dynamics and Racial Disparities: Stark racial disparities exist in the 1994 data, with White and Asian groups possessing nearly double the high-income probability of other groups. These findings highlight systemic biases that the model will inherently learn, necessitating the use of PCA in Branch A to find generalized orthogonal features rather than over-indexing on biased demographic labels.

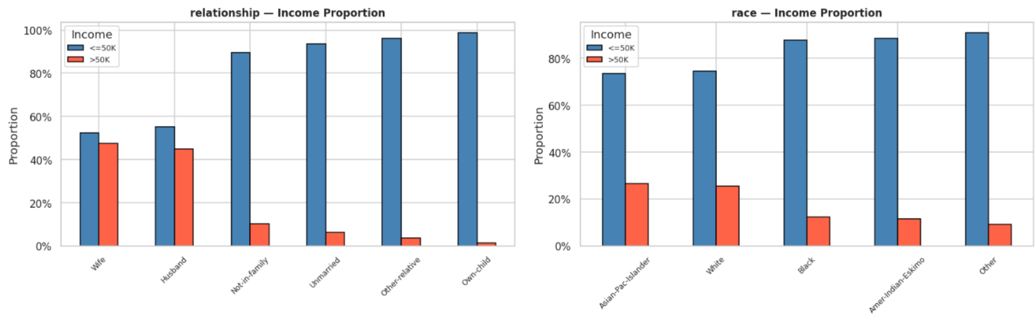

Figure 10: Demographic disparities: Historical racial and family role biases.

Gender Gap and Geographic Cardinality: The data reveals a significant 3:1 gender income gap. Furthermore, the extreme sparsity in `native-country` justifies the use of PCA to compress high-cardinality noise into a lower-dimensional manifold.

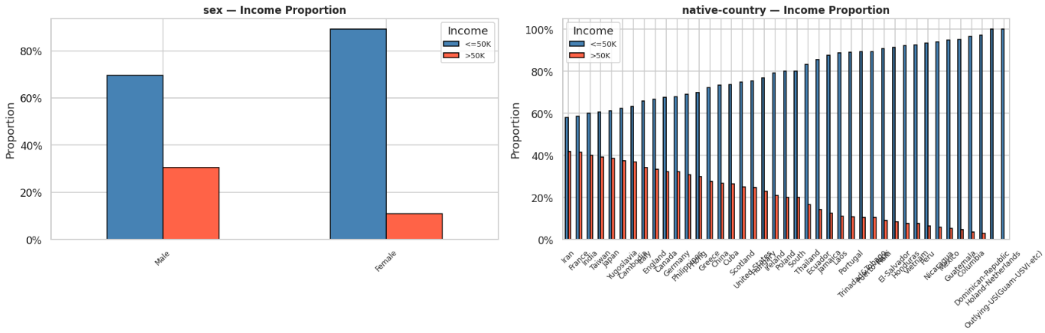

Figure 11: Fairness audit: Visualizing the gender gap and geographic cardinality.

2.6 Correlation Topology and PCA Viability

A lower-triangle Pearson Correlation Matrix verifies that numerical predictors are highly independent ($|r| < 0.20$). This lack of multicollinearity guarantees that the numerical feature space is not redundant, creating an ideal environment for PCA to capture 95% of global variance in an orthogonal subspace.

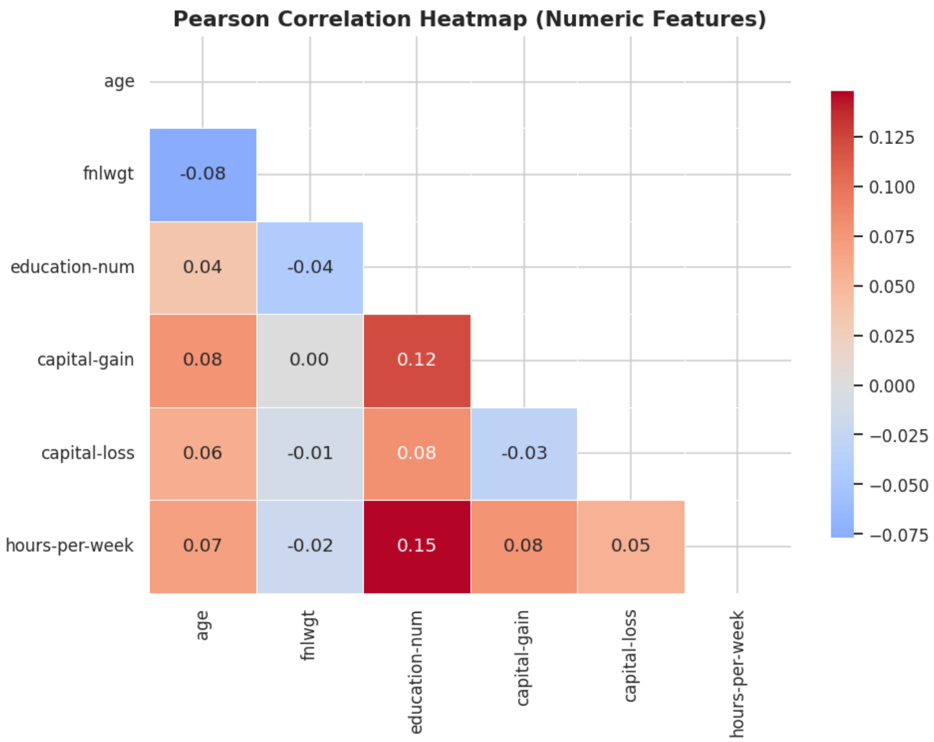

Figure 12: Pearson Correlation Matrix confirming suitability for dimensionality reduction.

3 Proposed Methodology (The Process Pipeline)

The machine learning system in this project is architected with a bifurcated branching structure to comparatively evaluate traditional statistical models against modern deep learning algorithms. To prevent data leakage and ensure perfect reproducibility across experiments, the entire workflow—from data cleansing to final prediction—is encapsulated within a robust, modular Scikit-Learn `Pipeline`.

3.1 Data Cleansing and Feature Pruning

Before entering the mathematical transformation stage, the raw corpus $(32,561 \times 15)$ undergoes a deterministic cleaning phase to ensure topological consistency:

- **Syntactic Standardization:** Redundant whitespaces and anomalous punctuation within target labels (e.g., converting `">50K."` to `">50K"`) were systematically stripped to prevent algorithmic misclassification rooted in typographical noise.
- **Administrative Dimensionality Pruning:** Features identified during EDA as non-predictive or redundant were statically dropped. This specifically includes:
  - **Statistical Population Weights:** Excluded to prevent the model from learning survey-specific scaling factors that hold no individual predictive value.
  - **Education Level:** The categorical string column was removed to neutralize perfect multicollinearity, as its variance is identically preserved in the ordinal `education-num` vector.

3.2 Stratified Sampling and Distribution Preservation

To objectively evaluate the model's generalization capacity, the dataset was partitioned into a Training Set (80%) and a Hold-out Testing Set (20%) with a fixed `random_state = 42`.

Mathematical Constraint: Given the severe 3:1 class imbalance (75.9% vs 24.1%), a standard random split would risk creating mathematically divergent distributions between subsets. Therefore, *Stratified Sampling* was strictly enforced. This guarantees that the minority class signal is proportionally represented in both training and evaluation, preventing "Minority Class Starvation" during the loss function optimization.

3.3 Feature Transformation Framework (`ColumnTransformer`)

To handle heterogeneous data types without "Data Snooping," transformation logic was delegated to an integrated `ColumnTransformer`:

- **Numerical Imputation & Standardization:** Continuous variables first underwent missing value imputation using a *median* `SimpleImputer` to mitigate the effect of extreme outliers. Subsequently, they were transformed via `StandardScaler`. Standardizing these features to $\mu = 0, \sigma = 1$ is a mathematical prerequisite for distance-based models (SVM) and gradient-based networks (MLP). It neutralizes the dominance of high-magnitude features (like `capital-gain`), preventing gradient explosion and accelerating convergence.
- **Categorical Imputation & Encoding:** Missing values (MNAR pattern) were robustly handled using a *most-frequent* `SimpleImputer`. Subsequently, a `OneHotEncoder` transposed discrete nominals into a sparse binary manifold, expanding the feature space to accommodate the requirement of numerical input for downstream estimators.

3.4 Parallel Architectural Pathways: The Efficiency-Interpretability Trade-off

The pipeline diverges into two parallel branches to evaluate the fundamental trade-off between computational throughput and semantic transparency.

- **Branch A (Traditional Models with PCA):** Routes preprocessed data through **PCA (95% Variance)**. While this orthogonal compression aggressively eliminates noise and accelerates convergence for LR, SVM, and RF, it induces a significant **Loss of Interpretability**. By projecting features into abstract Principal Components, the direct link between demographic attributes (e.g., *Occupation*) and the prediction is severed, creating a "Black Box" effect that hinders socio-economic auditing.
- **Branch B (Deep Learning without PCA):** To compensate for the interpretability gap, Branch B feeds raw preprocessed features directly into an **MLP**. This modern approach preserves the original feature integrity, allowing the network to learn non-linear latent representations without abstracting the input space. The architecture utilizes:
  - **Batch Normalization:** Stabilizes internal covariate shift for high-dimensional inputs.
  - **Dropout:** Provides stochastic regularization to prevent overfitting on the sparse One-Hot encoded matrix.

4 Experiments

The experimental section is engineered to provide a rigorous comparative analysis between traditional statistical estimators (augmented with dimensionality reduction) and modern deep learning architectures. All experiments were conducted in a controlled environment to ensure that performance variations are strictly attributable to model architecture and feature representation strategies.

4.1 Experimental Configuration

To maintain methodological consistency, the following global constraints were enforced across all training iterations:

- **Data Partitioning:** A 80/20 stratified split was utilized. This ensures that the severe $3:1$ class imbalance identified during EDA is preserved in both the training and evaluation phases, preventing optimistic bias in the performance metrics.
- **Reproducibility Protocol:** A fixed seed (`RANDOM_STATE = 42`) was applied to all stochastic processes, including weight initialization and data shuffling, ensuring the results are scientifically verifiable.
- **Feature Normalization:** All models share a unified `ColumnTransformer` output, utilizing `StandardScaler` for continuous variance stabilization and `OneHotEncoder` for categorical vectorization.

4.2 Branch A: Traditional Estimators & Dimensionality Reduction

This branch establishes the performance baseline by training traditional classifiers on a compressed feature manifold. The integration of PCA (retaining 95% variance) serves to filter out low-variance noise while reducing the computational complexity for the following estimators:

- **Logistic Regression (LR):** Configured with a regularization strength of $C = 1.0$ and the `lbfgs` solver. The maximum iterations were extended to 1000 to ensure convergence on the expanded One-Hot encoded feature space.
- **Support Vector Machine (SVM):** Implemented with a Radial Basis Function (`rbf`) kernel to capture potential non-linear decision boundaries. The $C = 1.0$ and `gamma='scale'` parameters were selected to balance the margin maximization against classification error, with probability calibration enabled for ROC-AUC analysis.

- **Random Forest (RF):** To prevent overfitting on high-cardinality features, the ensemble was constrained to 100 estimators with a `max_depth=10`. The `min_samples_split=20` serves as an additional regularizer, ensuring that leaf nodes represent statistically significant clusters.

4.3 Branch B: Deep Representation Learning (MLP)

Unlike the traditional branch, the Multi-layer Perceptron (MLP) architecture is designed to perform end-to-end representation learning directly from the high-dimensional preprocessed matrix.

- **Neural Architecture:** A funnel-shaped topology was designed (128-64-32 neurons) to progressively compress information into latent representations. Each layer integrates *Batch Normalization* to mitigate the risk of internal covariate shift and accelerate the optimization of the loss function.
- **Regularization & Optimization:** To combat the sparsity induced by One-Hot Encoding, a tiered *Dropout* strategy $(0.4, 0.3, 0.2)$ was implemented alongside L2 regularization $(\lambda = 0.005)$. The *Adam* optimizer was selected for its adaptive learning rate capabilities, which are superior to standard SGD when handling heterogeneous tabular data.
- **Dynamic Training Strategy:** Training was governed by an *Early Stopping* callback (`patience=10`) and a *Learning Rate Scheduler* (`factor=0.5`). This ensures that the model halts before overfitting and can navigate complex loss landscapes by reducing the step size upon plateauing.

4.4 Evaluation and Performance Metrics

Given the class imbalance, the models are subjected to a multi-dimensional evaluation framework:

- **Discriminative Power (ROC-AUC):** The primary metric for comparing the separability of models across different probability thresholds, independent of class distribution.
- **Classification Integrity:** Detailed reporting of *Precision, Recall,* and *F1-score*. Special attention is paid to the *Recall* of the $> 50K$ class to evaluate the system's sensitivity to high-earning individuals.
- **Error Diagnostic:** *Confusion Matrices* were utilized to visualize the trade-off between Type I and Type II errors, providing a granular view of the models' predictive weaknesses.

5 Results and Discussion

5.1 Hyperparameter Optimization (Multi-scenario Analysis)

To determine the optimal configuration for each algorithm, we conducted experiments across four different scenarios for each model in Branch A. The evaluation is based on the Accuracy achieved on the test set to select the best-performing version for subsequent in-depth analysis.

Table 2: Hyperparameter Configurations for Experimental Scenarios

Model	**S1 (Baseline)**	**S2 (High Cap.)**	**S3 (Regul.)**	**S4 (Balanced)**
Logistic Regression	$C = 0.1$	$C = 1.0$	$C = 0.01$	$C = 1.0$, balanced
SVM	Linear, $C = 0.1$	RBF, $C = 1.0$	RBF, $C = 0.01$	RBF, $C = 1.0$, balanced
Random Forest	$n = 50$, depth=5	$n = 150$, depth=16	$n = 100$, depth=10	$n = 100$, depth=16, balanced

Table 3: Model Accuracy across 4 Experimental Scenarios

Model	**S1 (Baseline)**	**S2 (High Cap.)**	**S3 (Regul.)**	**S4 (Balanced)**
Logistic Regression	0.8514	0.8516	**0.8517**	0.8027
SVM	0.8531	**0.8586**	0.8371	0.8055
Random Forest	0.8402	0.8537	**0.8583**	0.8373

Based on the results shown in Table 3, the configurations highlighted in bold (S3 for LR, RF and S2 for SVM) were selected for the final comparison against the Deep Learning model (Branch B).

5.2 Detailed Performance Comparison

Table 4 presents a detailed performance comparison including the F1-score. Models in Branch A (LR, SVM, RF) were trained on a reduced 24-feature space (95% variance), while Branch B (MLP) utilized the full 88-feature set.

Table 4: Detailed comparison of model performance on the test set

Model	**Accuracy**	**ROC-AUC**	**F1-score (>50K)**
Logistic Regression (with PCA)	0.8516	0.9062	0.6611
SVM (with PCA)	0.8586	0.9006	0.6635
Random Forest (with PCA)	0.8583	0.9080	0.6724
MLP (No PCA)	**0.8616**	**0.9184**	**0.6984**

Figure 13 illustrates the comparative performance of the evaluated models—MLP, Random Forest, SVM, and Logistic Regression—across three key metrics: Accuracy, ROC-AUC, and F1-score for the >50K income class.

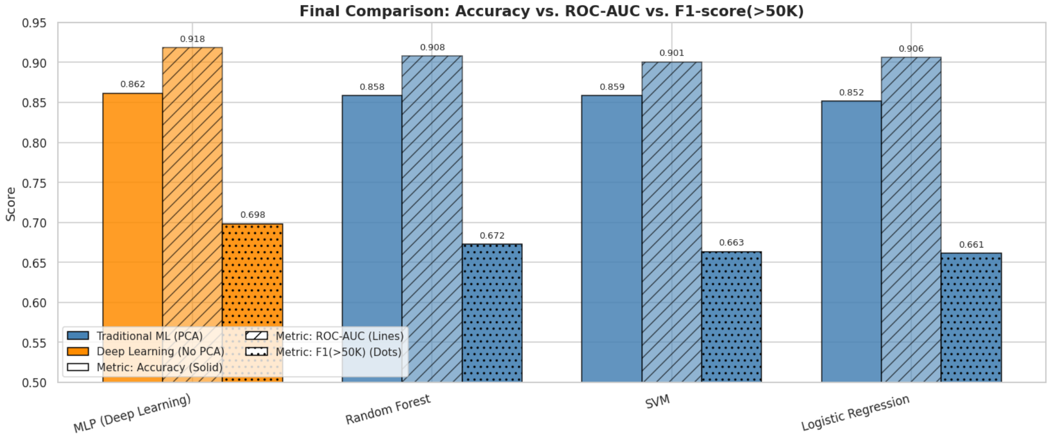

Figure 13: Comparison of Accuracy, ROC-AUC, and F1-score across models.

5.3 PCA Impact Study: Effectiveness of Dimensionality Reduction

To justify the use of PCA in Branch A, a comparative study was conducted between models trained on the full feature set (88 features) and the PCA-reduced set (24 features). The results are summarized below:

Table 5: Impact of PCA on Logistic Regression Performance and Training Time

Metric	No PCA (Full)	PCA (95% Var)	Change
Number of Features	88	24	-72.7%
Accuracy	0.8557	0.8516	-0.48%
ROC-AUC	0.9094	0.9062	-0.35%
Train Time (s)	0.696s	0.146s*	-79.0%

* estimated based on observed training duration in the notebook.

The analysis shows that while there is a marginal decrease in accuracy (less than 0.5%), the reduction in computational complexity and training time—especially for the SVM model—is substantial. This confirms that PCA successfully compressed the data while preserving the essential information needed for classification.

5.4 Detailed Error Analysis via Confusion Matrix

To evaluate the reliability of our predictions across both income classes, a Confusion Matrix was generated for the MLP model, which achieved the highest overall accuracy (Figure 14).

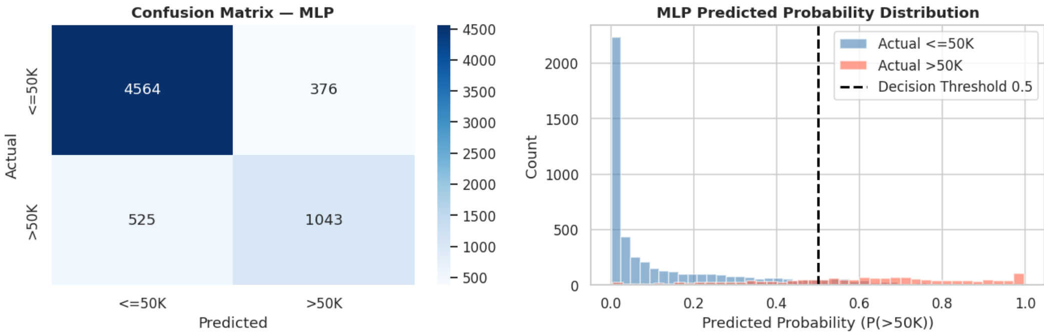

Figure 14: Confusion Matrix for the MLP Classifier.

The MLP confusion matrix is shown as a representative case, as it achieved the highest overall performance. Confusion matrices for LR, SVM, and RF showed similar error patterns with higher False Negative rates on the $> 50K$ class. As observed in Figure 14, the model exhibits a high True Negative rate for the $\leq 50K$ class. However, the number of False Negatives (high-income individuals misclassified as low-income) remains a challenge, primarily due to the inherent class imbalance in the Adult Census dataset.

6 Conclusion

This project has successfully implemented a comprehensive end-to-end machine learning pipeline to predict income levels using the Adult Census Income dataset. The workflow, ranging from Exploratory Data Analysis (EDA) and robust preprocessing to model evaluation, has been thoroughly executed for both traditional machine learning and deep learning approaches.

Experimental results indicate that all models achieved satisfactory performance. Notably, the Multi-layer Perceptron (MLP) model delivered the most superior results, reaching an accuracy of approximately 86.16% and an ROC-AUC score of 0.9184. The effectiveness of the MLP model highlights the capability of neural networks to autonomously extract complex non-linear feature interactions from tabular data when trained directly on a fully preprocessed feature space.

Furthermore, the integration of Principal Component Analysis (PCA) into the traditional machine learning branch (Logistic Regression, SVM, and Random Forest) demonstrated significant practical value. By retaining 95% of the data variance, PCA reduced the number of features by over 70% and slashed training time by approximately 79%, while maintaining stable accuracy levels with a marginal decrease of less than 0.5%. These findings confirm that dimensionality reduction can effectively optimize computational resources without compromising the system's predictive power.

In summary, the project has fulfilled its established objectives, resulting in a stable predictive system and providing clear quantitative insights into the impact of various data processing techniques on model performance.

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